

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027

Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

Culture & Morality Lab, UMass Amherst, Lab Assistant | *Amherst, MA* Jul 2026 – Present

- Sole engineer building the lab's text-analysis platform (Python, **FastAPI**, sentence-transformers, React), owning architecture, the production design doc, and all technical decisions end to end.
- Designed the systems backend, a DB-persisted job queue with orphaned-job recovery, additive schema auto-migration, and a versioned data registry (**99 constructs from 38 questionnaires**); grew the suite to **40 hermetic tests** running in CI without ML dependencies.

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clinical clients**, processing **10,000+ monthly triage submissions** via REST integrations; contributed to **\$340K+** in revenue.
- Owned day-to-day reliability of the multi-tenant service, monitoring production logs during releases and client integrations and shipping targeted fixes for tenant-specific incidents to keep it stable; collaborated with clinical and product stakeholders to turn requirements into backend features.
- Architected a full-stack ML annotation platform (React, Node.js, MongoDB) used daily by the AI team, cutting manual annotation by **100+ hours/month** and lifting labeling throughput by **60%**.

Cario Growth Services, Backend Developer Intern | *Chennai, India* Nov 2022 – May 2023

- Designed and implemented **20+ RESTful APIs** (Node.js, Fastify, PostgreSQL) with input validation, clear contracts, and backward-compatible changes used in production by the frontend team.
- Shipped a production LLM feature on **4-bit quantized Llama-7B via llama.cpp** on a single VM, with async queued generation, per-character rate limits, short-circuit caching, and a deterministic content-safety filter, built end to end with no ML mentor.

Projects

Cache- and SIMD-Aware Matrix Multiplication (GitHub) C++, AVX/SIMD, OpenMP, Linux perf

- Optimized C++ matrix multiplication from a naive baseline through cache-aware tiling, manual **AVX/SIMD** vectorization via `__m256` intrinsics, register-aware micro-kernels, and **OpenMP** multithreading, profiling L1/L3/TLB misses with perf and inspecting generated assembly.

Spark ETL Performance Optimization (GitHub) PySpark, Spark UI, SQL, NYC TLC Dataset

- Cut end-to-end Spark ETL runtime **~25%** (17–19 min to 14.3 min) on a **~1.4B-row** dataset via broadcast joins, Adaptive Query Execution, and skew-join handling, and reduced the analytics tail-latency ratio **2.14x to 1.74x** using Spark UI stage profiling to localize bottlenecks.

Sonare – Offline Sign / Speech App (GitHub) React, Electron, FastAPI, MediaPipe, whisper.cpp

- Engineered async APIs and job queues to keep the UI responsive while concurrently handling local video and audio streams in a fully offline, on-device pipeline (MediaPipe, **whisper.cpp**, local TTS) with no external network dependencies.

Publications

- "Alzheimer's Disease Classification using Transfer Learning," IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages: C++, Python, TypeScript, JavaScript, Java, SQL, Bash

Systems & Performance: AVX/SIMD (`__m256`), OpenMP, multithreading, cache-aware tiling, Linux perf, Cachegrind, TMA profiling

Concurrency & Backend: async services & queues, REST APIs, FastAPI, Node.js, Express, distributed systems

Data & Databases: PySpark, Spark UI, distributed data processing, PostgreSQL, MongoDB, Redis, SQLite

Cloud & DevOps: AWS, GCP, Azure, Docker, Git, GitHub Actions, Linux, CI/CD